

BST Physics Curriculum Vol 10 Chapter 8 — Representation Theory v0.4 (refilled per Cal #104)

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Vol 10 Chapter 8 — Representation Theory

Chapter motivation

Standard graduate representation theory: representations of finite groups (Maschke + Schur lemma + character theory + irreducible reps); compact Lie group reps (Peter-Weyl theorem + Weyl character formula + classification by highest weights); non-compact + semisimple Lie group reps (Harish-Chandra). Standard texts: Fulton-Harris + Knapp + Hall.

BST cross-link: Wallach 1976 classification of holomorphic discrete series for bounded HSDs gives K-type decomposition of Bergman $H^2(D_{IV}^5)$ (Vol 11 Ch 3). K-types $V_{(p,q)}$ under $K = SO(5) \times SO(2)$ play role of standard Fourier modes (Vol 10 Ch 5 cross-link); Casimir spectrum (Vol 1 Ch 5) gives substrate physics observables.

Section 8.0b — Reader-grade 3-level pedagogy

Level 1: Standard finite + compact group representation theory + classification by highest weights; BST cross-link: Wallach 1976 K-type classification of Bergman $H^2(D_{IV}^5)$ is substrate-Fourier framework.

Level 2 (graduate-physicist): Standard rep theory: finite groups G have finitely many irreducible reps; Maschke (over char 0) — every rep decomposes as direct sum of irreps; Schur lemma — endomorphisms of irrep are scalars; character orthogonality + Burnside count of irreps. Compact Lie groups ($SU(n)$, $SO(n)$, etc.): Peter-Weyl theorem — $L^2(G) = \bigoplus_{\rho} V_{-\rho} \otimes V_{\rho}^*$ over irreps ρ ; Weyl character formula gives explicit character of each irrep; classification by highest weights (dominant + integral). Non-compact semisimple Lie groups ($SL(n, \mathbb{R})$, $SO_0(p, q)$, etc.): Harish-Chandra theory; discrete series + principal series + supplementary series; intertwining operators. BST substrate cross-link: substrate group $SO_0(5, 2)$ (Vol 10 Ch 6) acts

on Bergman $H^2(D_{IV}^5)$ via unitary representation. Restriction to $K = SO(5) \times SO(2)$: Bergman H^2 decomposes per Wallach 1976 (Vol 11 Ch 3) as $\oplus V(p,q)$ over admissible (p, q) — holomorphic discrete series of $SO_0(5,2)$ is the Bergman space; K-type $V_-(p,q)$ is the (p, q) -isotypic component. Casimir eigenvalue $Cas(p, q)$ per K-type gives substrate physics observable spectrum (Vol 1 Ch 5). Ground state $V_-(0,0)$ with Casimir = $C_-2 = 6$ BST primary (T2439 Strong-Uniqueness C4 RIGOROUSLY CLOSED). All BST physics observables = evaluations on K-type basis (substrate-Fourier per Vol 10 Ch 5).

Level 3 (5th-grader accessible): Representation theory studies how groups act on vector spaces. BST's substrate group $SO_0(5,2)$ acts on Bergman $H^2(D_{IV}^5)$; the decomposition into Wallach K-types $V_-(p,q)$ IS substrate-Fourier (Vol 10 Ch 5 cross-link). Ground-state K-type has Casimir eigenvalue = $6 = C_-2$ (BST integer).

Section 8.1-8.4

Standard finite group + compact Lie group + non-compact semisimple Lie group representation theory; BST Wallach K-type cross-link.

Bergman $H^2(D_{IV}^5) = \oplus V_-(p,q)$ over admissible Wallach K-types; Casimir $Cas(p, q)$ eigenvalue spectrum gives substrate physics.

Section 8.5 — Honest scope + Connection

- Standard representation theory [\[?\]](#)
- Wallach K-type substrate cross-link [\[?\]](#)
- Vol 11 Ch 3 detailed treatment cross-reference
- Vol 1 Ch 5 Casimir algebra (substrate physics observables)

— Lyra, Vol 10 Ch 8 v0.4 chapter-grade narrative REFILLED per Cal #104, Saturday 2026-05-23 EDT